Phase 1 & 2 FortiGate WAN Connectivity over MPLS and ISP Infrastructure

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FortiGate - Internet WAN Configuration

A FortiGate firewall utilizes a standard public internet connection paired with static routing to establish a reliable, fixed path for outbound web traffic and backup corporate connections. By configuring static routes with a higher Administrative Distance (like 200), the firewall ensures this path stays quietly in the background as an inactive standby route while your primary MPLS link is healthy. When the primary link goes down, the FortiGate instantly brings these static routes into the active routing table, immediately building an IPsec VPN tunnel across the internet to reconnect your branches. This architecture gives you a highly cost-effective, securely encrypted failover solution that keeps business operations online during a major circuit outage.

What it is

This section simulates customer internet connectivity through ISP routers using OSPF dynamic routing & static routing through ISP and Fortinet firewall.

Device involved:

- HQ FortiGate
- ISP-1
- INTERNET Router
- ISP-2
- Branch FortiGate

Routing Used:

- Static Route on FortiGate
- OSPF between ISP routers

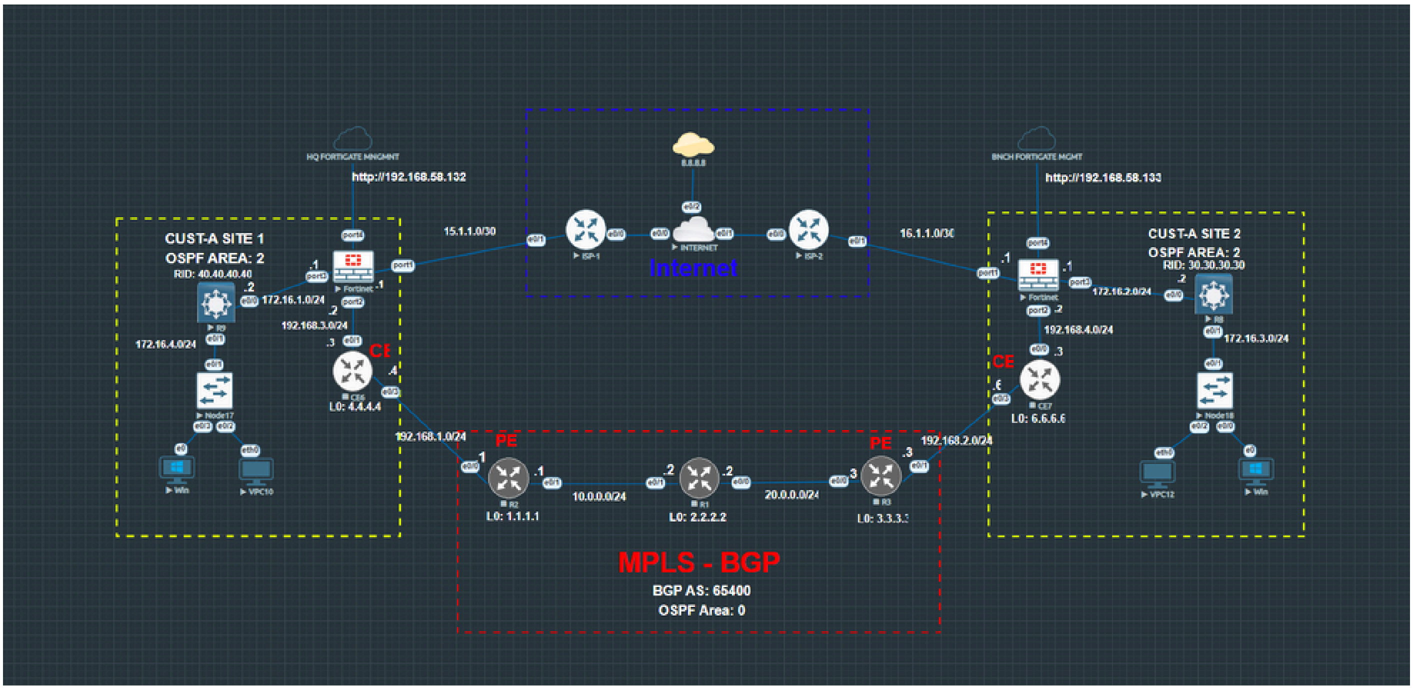
Why it exists

This setup simulates how customer firewalls connect to the public internet through an ISP provider.
The FortiGate firewall receives a WAN IP address and forwards traffic toward the ISP gateway using static routing. OSPF is used internally between ISP routers to dynamically exchange routes and maintain connectivity across the simulated internet.

How it works together

Traffic Flow:

- HQ FortiGate → ISP-1 → INTERNET → ISP-2 → Branch FortiGate
- The FortiGate firewalls use static routes toward their ISP gateways.
- The ISP routers dynamically exchange routes using OSPF Area 99.
- This allows both customer firewalls to communicate successfully across the internet network.



Result

- OSPF neighbors established successfully
- ISP route exchange operational
- End-to-end internet communication is successful
- 0% packet loss during ping tests

Note

- Same Configurations apply to the other branch just replace the Subnets

FortiGate - MPLS-BGP PROVIDER NETWORK Configuration

A FortiGate firewall sits at the edge of your network to provide deep security inspection for all data entering or leaving the private MPLS cloud, which natively lacks threat filtering. It runs dynamic routing protocols like OSPF to smoothly exchange network paths with the service provider's routers. Using advanced SD-WAN features, the FortiGate constantly measures the health of the MPLS circuit to ensure optimal performance for your critical corporate applications. If that private line experiences an outage, the firewall instantly shifts traffic over to your backup internet VPN tunnel without disrupting the business.

What it is

This section simulates an MPLS service provider network using:

Device involved:

- CE Routers
- PE Routers
- P Router
- OSPF
- MPLS LDP
- MP-BGP

Routing Used:

- CE6
- PE1 (R2)
- P Router (R1)
- PE2 (R3)
- CE7
- Customer FortiGate Firewalls

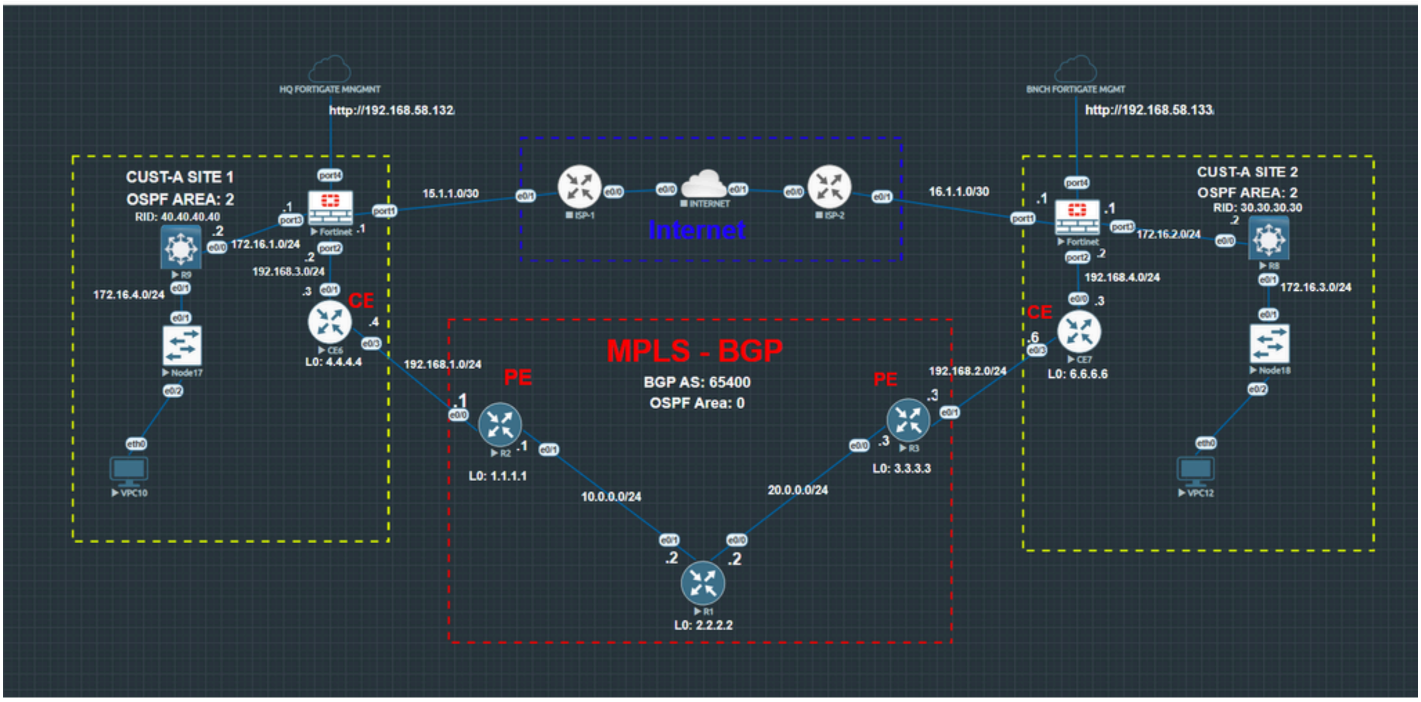
Why it exists

The MPLS network allows customer branches to communicate privately through the provider backbone without using the public internet. This simulates real enterprise WAN services provided by telecommunications companies. OSPF is used for internal provider routing. MPLS distributes labels across the core. MP-BGP exchanges customer VPN routes between PE routers.

How it works together

Traffic Flow:

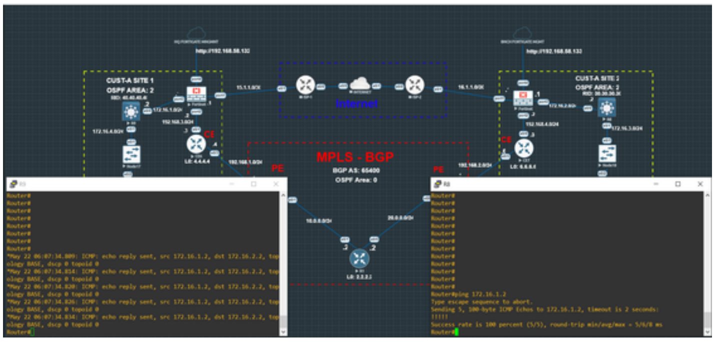
- FortiGate → CE → PE → P → PE → CE → FortiGate
- CE routers connect customer networks toward the provider edge.
- PE routers exchange customer routes using MP-BGP.
- The P router acts as the MPLS core transport router.
- OSPF provides internal reachability while MPLS labels forward traffic efficiently through the provider network.

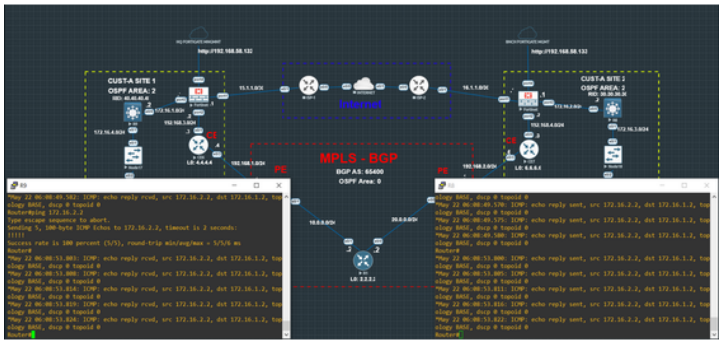


Result

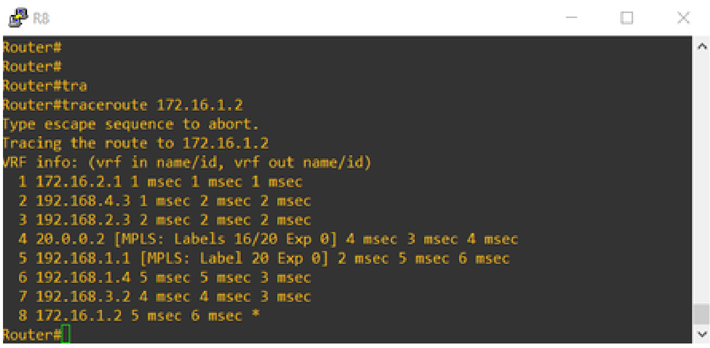
- MPLS labels exchanged successfully
- BGP neighbors established successfully
- OSPF adjacencies operational
- End-to-end MPLS communication successful
- Firewall-to-firewall connectivity verified
- 0% packet loss during testing

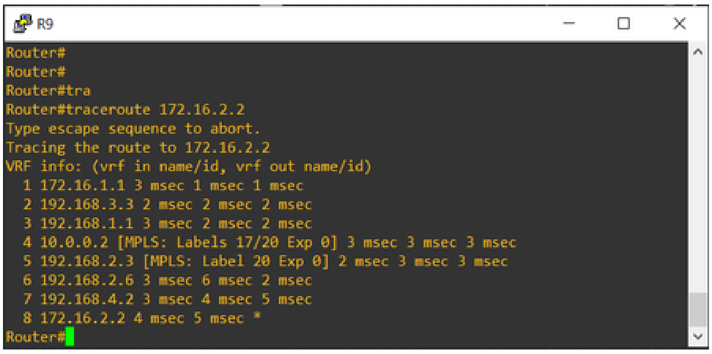
PING TEST ON LAN ROUTERS



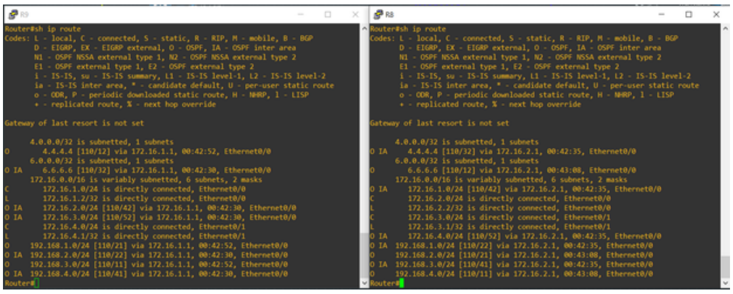


TRACEROUTE

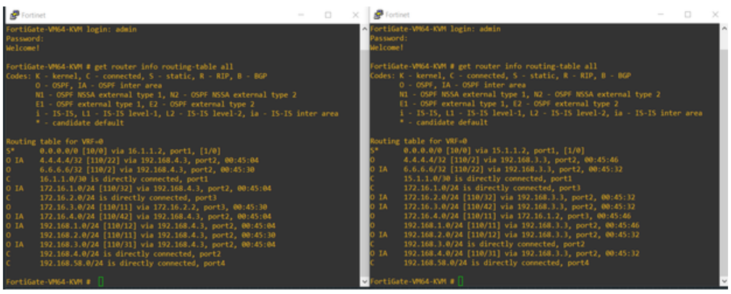




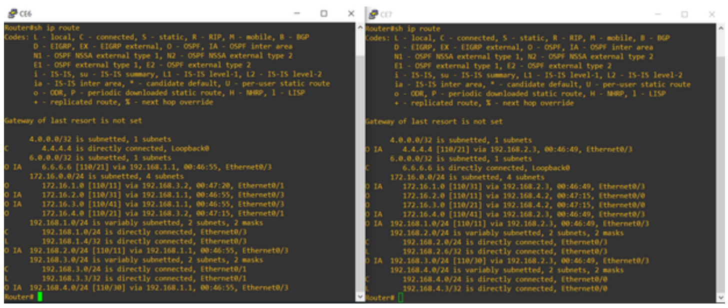
ROUTING TABLE FOR TROUBLESHOOTING



LAN ROUTERS ROUTING TABLE



FIREWALL ROUTERS ROUTING TABLE



CE ROUTERS ROUTING TABLE

Conclusion

THIS PROJECT DEMONSTRATED:

- Internet WAN routing using OSPF
- MPLS provider core architecture
- MP-BGP route exchange
- CE/PE/P router deployment
- FortiGate WAN integration
- Dynamic routing implementation
- End-to-end branch communication

The lab simulates a real-world enterprise WAN environment commonly used by service providers and large enterprise organizations.

THE PROJECT ALSO IMPROVED PRACTICAL SKILLS IN:

- Routing and Switching
- MPLS Technologies
- BGP Configuration
- OSPF Troubleshooting
- WAN Connectivity
- FortiGate Administration
- Network Verification and Testing

Note

- Same Configurations apply to the other branch just replace the Subnets